CUSTOMER SUPPORT PILOT
Report

Q1: What worked, and what failed?

All three versions achieved 100% refusal accuracy and zero false refusals. Retrieval metrics improved from V1 to V2 advanced and multi-index with routing was strongest overall: Precision-38.75%, Recall and multi-source coverage 100%, routing accuracy 100%, and RAGAS faithfulness 0.9767.

The main weakness is Precision, even the best version retrieves some top-five documents that are not counted among the expected source IDs. Answer relevancy is moderate (0.7109-0.7412), and question-level checks exposed omissions: the Q03 demo should also say express delivery is unavailable for remote pin codes, while Q01 should explicitly give the current policy's effective date.

Q2: One real bug you found, and how you fixed it

One real bug which I faced was RAGAS evaluation crashed by showing IncompleteOutputException. During generation evaluation, the script repeatedly crashed while calculating RAGAS faithfulness for Q06 in the V2 multi-index pipeline. RAGAS was unable to parse the model's structured response completely, preventing the remaining questions from being evaluated.

As to how I fixed the bug was by modifying the maxtokens count parameter.

Q3: In your V2 pipeline (Section 4), what order did you put the steps in, and why?

Steps in order:

1. Query transformation
2. Metadata filtering
3. Dense and BM25 retrieval
4. RRF fusion
5. Cross-Encoder up to top5
6. Answer generation with the sources

I followed this order to improve the retrieval quality before generating the final answer. Query transformation helps express the user's question in easier way. The hybrid search combines dense vector search which finds semantical similar content with BM25 keyword search thru which helps to match exact terms with the parameters such as products names and the policy ids. RRF combines the ranked results. The cross-encoder re-ranks the top5.

Finally, the model generates an answer using the retrieved context and provided source ids.

Q4: Each record was kept as one chunk. Why was extra splitting not required for this dataset?

Each record was kept as one chunk because the 50 records in the Dandes electronic dataset the base was already short and self-contained. Each record had all the relevant data.
Further splitting was unnecessary because the records were small enough and splitting then again would separate all the relevant documents which were together.

Q5: Single index vs multi-index: which questions improved, and which got worse?

The Multi-Index RAG system with intent-based routing improved overall retrieval performance compared with the V2 Single-Index system. Precision increased from 35.00% to 38.75%, Recall increased from 95.42% to 100%, and multi-source coverage improved from 92.67% to 100%.
Questions that improved:

- Q06 — Damaged phone refund: Multi-Index provided both the official process and the resolved case, TCK-1002. Answer relevancy improved from 0.674 to 0.718, and context recall increased from 0.833 to 1.000.
- Q08 — Galaxy Buds3 Pro troubleshooting: RAGAS faithfulness increased from 0.600 to 0.833, while answer relevancy improved from 0.621 to 0.709.
- Q14 — Delayed Cash on Delivery refund: RAGAS faithfulness improved from 0.333 to 1.000, and context precision increased from 0.750 to 0.833.
- Q18 — Product compatibility: The Multi-Index answer addressed all three compatibility questions, whereas the Single-Index answer omitted parts of the question. Answer relevancy improved from 0.721 to 0.907.

Questions that got worse:

- Q04 — Changing an address after shipment: Answer relevancy decreased from 0.858 to 0.697, although the multi-index answer still covered the procedure and resolved case.
- Q12 — Defect after the return window: Answer relevancy decreased from 0.775 to 0.740.
- Q15 — Defective replacement: Answer relevancy decreased from 0.624 to 0.512, despite context recall improving from 0.667 to 1.000.
- Q16 — Customer moved after shipment: Answer relevancy decreased from 0.685 to 0.531, although the answer included the relevant procedure and resolved example.
- Q01 — Opened laptop return: RAGAS context recall decreased from 0.667 to 0.333, indicating that the retrieved context was assessed as covering less of the reference answer on this question.

Q6: One question where the router picked the wrong index: why it happened, and how you fixed it.

In Q12, the question was: “A product's return window has ended, but it has developed a defect. What should the customer do?”

The expected index was order_index, but the router selected both policy_index and order_index. This likely happened because the phrase “return window” triggered the policy intent, even though the question primarily concerned the warranty procedure for a defective product. The required order_index was still selected, so the extra index did not prevent the system from retrieving the necessary information. Under the assignment's routing evaluation rule, selecting all expected indexes with at most one extra index counts as correct; therefore, the final routing accuracy remained 100%.

I identified this as an opportunity to improve routing by refining the intent instructions to prioritize order_index for warranty claims after the return period has ended, while using policy_index when the question requires specific return or refund rules.

Q7: How did your copilot handle documents that disagree with each other?

The copilot prioritizes the latest applicable policy over older information when documents conflict. I used the document dates and policy information in the knowledge base to distinguish current rules from older resolved support tickets.

For example, an older support ticket mentions a seven-day return window for opened laptops, while the current policy, RR-001, effective 1 July 2026, specifies a ten-day return window. The copilot follows the current policy and explains that the older information has been superseded. It also cites the relevant policy sources, RR-001 and RR-002.

This helps prevent outdated information from being presented as the current rule. However, the system still depends on retrieving the relevant documents and recognizing the date difference, so handling conflicts remains an area to test carefully.

Q8: Which metric improved the most from V1 to V2, and which change caused it?

The metric that changed from v1 to v2 is multi-source coverage from 87.67% in v1-basic rag to 92.67% in v2-advanced rag.

This improvement was basically due to use of advanced retrieval techniques in v2. Usage of metadata filtering, hybrid search, dense retrieval and BM25, RRF also improved it.

Q9: Your lowest-scoring question: was it a retrieval problem or a generation problem?

My lowest scoring question was Q15- A replacement product is also defective question with answer relevancy score of 0.512.

It was a generation problem cause the required info was retrieved but the answer was not focused on the question.